CAPSTONE PROJECT · REAL ESTATE / INVESTMENT ANALYTICS

Real Estate Investment Advisor

Predicting property profitability and future value: a classifier for **Good Investment** and a regressor for the **price in 5 years**, tracked with MLflow and served through a Streamlit app.

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Date	17 October 2026
GitHub	https://github.com/aryaman2213/real-estate-investment-advisor
Stack	Python, pandas, scikit-learn, XGBoost, MLflow, Streamlit, Plotly

250,000
listings, 42 cities, 20 states

0.970
classifier ROC AUC · XGBoost (tuned)

14.9 L
regressor RMSE (R^2 0.9950) · Linear Regression

14
models trained and logged in MLflow

1. Problem statement and business use cases

Build a machine-learning application that helps investors decide (1) whether a property is a **Good Investment** (classification) and (2) what it will be worth **after 5 years** (regression), with experiment tracking in MLflow and an interactive Streamlit app.

- **Investors** assess long-term returns before buying.
- **Buyers** find high-return properties in developing areas (the app ranks cities and lists the top-scored properties).
- **Real estate platforms** automate investment analysis for every listing (the explorer scores all 250,000).
- **Trust**: every prediction shows its confidence, a forecast range and the reasons behind it.

2. Data and cleaning

`india_housing_prices.csv` : 250,000 rows × 23 columns (location, type, size, price, age, floors, nearby schools and hospitals, transport, parking, security, amenities, facing, owner, availability).

Check	Finding	Action
Missing values	None	Median / mode imputation kept in the pipeline as a safety net for app input
Duplicates	None (also none ignoring ID)	—
Floor number > total floors	116,304 rows (46.52%)	Values swapped back (column mix-up), no rows lost
Price_per_SqFt units	Stored in lakhs, rounded to 2 decimals	Recomputed exactly in rupees
Outliers (IQR)	Price 0, size 0, price per sq ft 19,723	Genuine small, expensive homes: flagged, kept; log price for linear models
Locality	Each of 500 values appears in all 42 cities	Excluded from models (no location information)
State / Year_Built	Determined by City / Age	Excluded as redundant

Key EDA finding: the listings are synthetic. Every raw column is uniformly distributed and independent of every other (size vs price r ≈ 0.00; average price per city varies by only 3%).

Feature engineering

`Price_per_SqFt` (₹), `Log_Price`, `PPSF_vs_City_Median`, `City_Tier` (8 X-class metros vs the rest), `Infrastructure_Score` (0–100: schools, hospitals, transport), `Amenity_Count`, `Amenity_Density_Score` (amenities per 1,000 sq ft), five `Has_<amenity>` flags, `Transport_Score`, `Furnished_Level`, `Floor_Ratio`, and binary parking / security / ready-to-move flags.

3. Target definitions

The dataset has no sale history, so both targets follow documented domain rules, as the brief allows.

> **Future_Price_5Y** = Price × $(1 + r)^5$, where the annual appreciation rate r = 5.5% base + feature adjustments + a random market shock (σ = 0.75 pp) for what a listing cannot capture. The market-wide average is ≈ 8%, the rate in the brief's example.

Driver	Adjustment (pp per year)	Reason
Public transport	Low -1.0, Medium 0, High +1.0	Connectivity drives resale demand
Under construction	+0.75	Pre-launch pricing; value rises on completion
Property type	Villa +0.5, Independent house +0.25	Land appreciates; buildings age
Tier 2 city	+0.75	Developing markets grow faster from a lower base
Security, parking	+0.25 each	Resale premium
Amenities; schools + hospitals	+0.25 each; +0.05 each	Facilities and social infrastructure
Age	-0.04 per year of age above 15 (newer homes gain the same per year below it)	Depreciation of older buildings

> **Good_Investment** = 1 when annual appreciation ≥ **8% and** price per sq ft ≤ the **city median**. This combines the brief's three suggested rules (appreciation above a threshold, cheaper than the median, multi-factor). Result: 27.06% of listings are Good Investments.

4. Exploratory data analysis

Price and size (Q1–Q5)

Q1. What is the distribution of property prices?

Prices are spread **evenly from ₹10 L to ₹500 L** (mean ₹254.6 L, median ₹253.9 L, skew ≈ 0). There is no long right tail like a real market would show: the distribution is uniform.

Q2. What is the distribution of property sizes?

Sizes are also **uniform between 500 and 5,000 sq ft** (mean ≈ median ≈ 2,750 sq ft, skew 0).

Q3. How does the price per sq ft vary by property type?

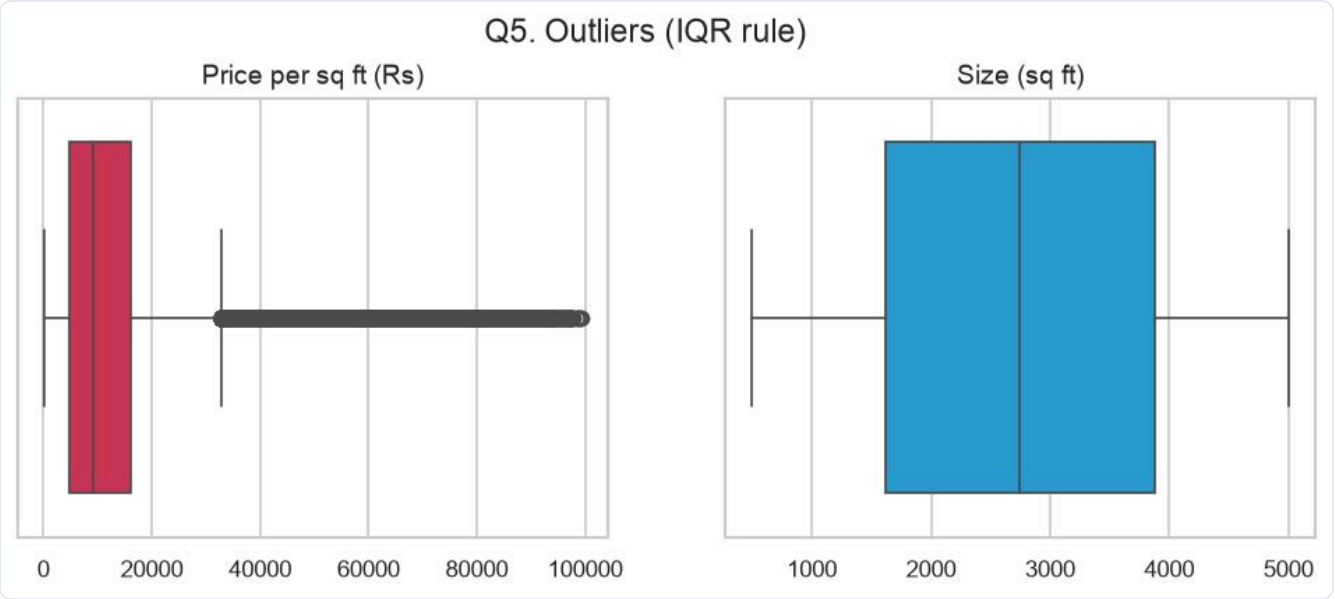
Medians are within about 1% of each other (~₹9,200–9,300/sq ft) for apartments, independent houses and villas. The Kruskal-Wallis p-value is small only because n = 250,000: the difference is **statistically detectable but practically meaningless**. Property type does not set price per sq ft in this data.

Q4. Is there a relationship between property size and price?

No. Pearson r ≈ 0.00 (p ≈ 0.2) and Spearman ρ ≈ 0: a 5,000 sq ft home is no more expensive on average than a 500 sq ft one. Because price and size are independent, **price per sq ft varies mostly with size**: big homes are much cheaper per sq ft.

Q5. Are there any outliers in price per sq ft or property size?

Size and price have **no outliers** (both bounded and uniform). Price per sq ft has **19,723 IQR outliers (7.9%)** above ₹32,764/sq ft, up to ₹99,182. They are genuine combinations (a small home at a high price), so they were **flagged (`Is_PPSF_Outlier`) but kept**; tree models are robust to them and linear models use `Log_Price`.



Q5. Outliers: only price per sq ft has them (IQR rule)

Location (Q6–Q10)

Q6. What is the average price per sq ft by state?

All 20 states fall in a narrow band, about **₹12,930–13,250/sq ft (±1.2%)**. Karnataka and Andhra Pradesh are highest, Punjab and Rajasthan lowest, but the gap is tiny.

Q7. What is the average property price by city?

City averages range only from about **₹250.8 L to ₹258.5 L (3%)**. Mumbai (₹252.1 L) is among the five cheapest cities on average: the raw data carries **no real location premium**. That is why location enters the targets through a documented city-tier rule rather than being learned from prices.

Q8. What is the median age of properties by locality?

Median age by locality ranges from **16 to 21 years** around an overall median of 18. Each locality name appears in all 42 cities, so locality is a label, not a neighbourhood.

Q9. How is BHK distributed across cities?

Every city has roughly **20% of each BHK (1–5)**, between 18.6% and 22.0% in every cell. There is no city skewed towards small or large homes.

Q10. What are the price trends for the top 5 most expensive localities?

The top 5 localities average ₹269–275 L. There are no sale dates, so construction year is the only time axis. Their prices by year built go up and down with no consistent trend: the differences are sampling noise, not a market trend.

Relationships (Q11–Q15)

Q11. How are numeric features correlated with each other?

The raw features are uncorrelated with each other (|r| < 0.01 everywhere). The structure is in the engineered features and targets: growth rises with transport (r ≈ 0.60) and infrastructure (≈ 0.53) and falls with age (≈ –0.29). `Good_Investment` is negatively related to price (≈ –0.40) and price per sq ft (≈ –0.38), and positively to size (≈ +0.30) and transport (≈ +0.30). Future price is almost entirely today's price (r ≈ 0.99).

Q12. How do nearby schools relate to price per sq ft?

No relationship (r ≈ 0.00, p ≈ 0.98). Median price per sq ft is ≈ ₹9,150–9,350 whether there is 1 school or 10.

Q13. How do nearby hospitals relate to price per sq ft?

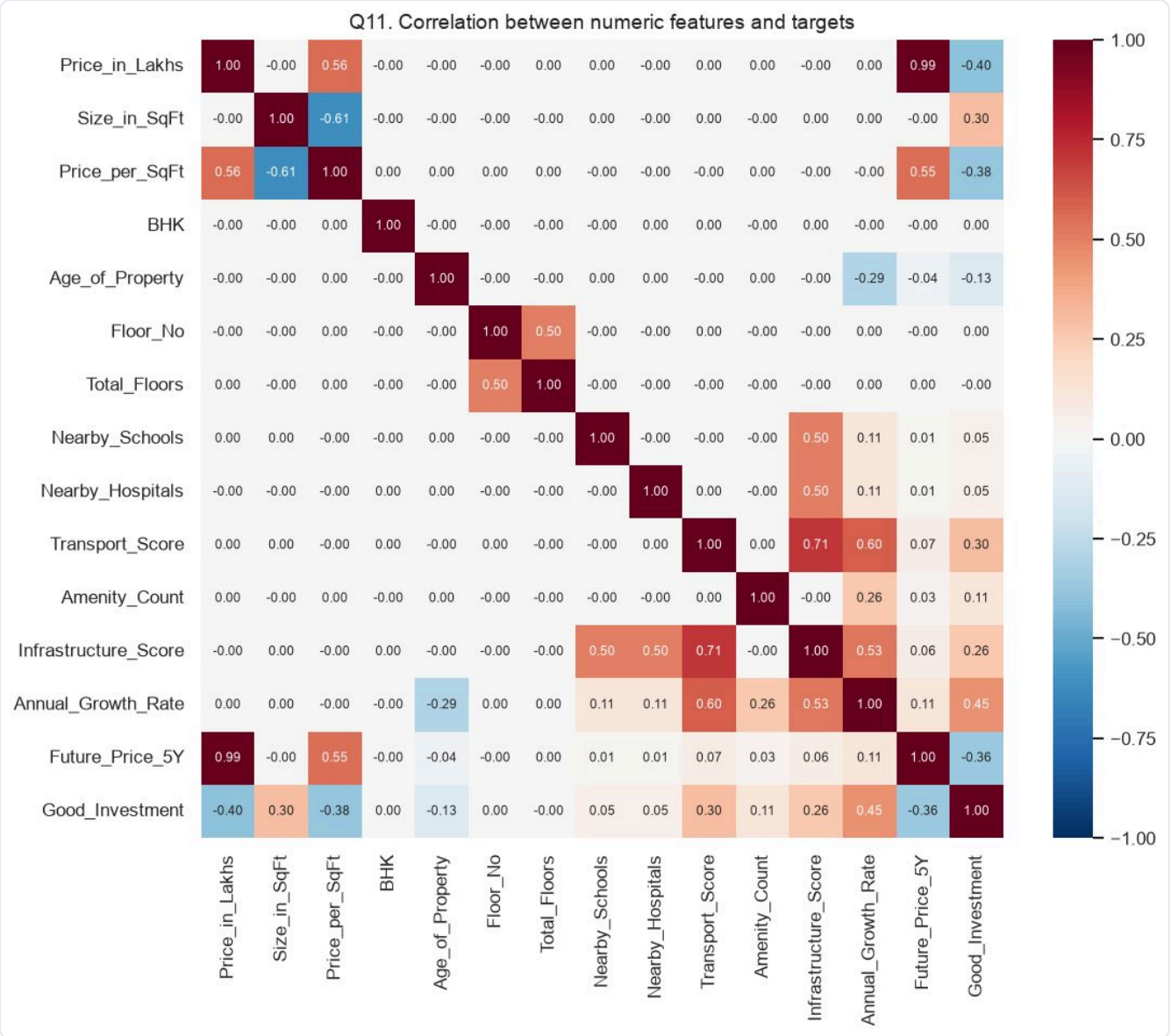
No relationship either (r ≈ 0.00, p ≈ 0.95). Schools and hospitals matter in this project through *appreciation* (Infrastructure Score), not today's price.

Q14. How does price vary by furnished status?

Average prices are ₹254.3–255.0 L for all three levels (ANOVA p ≈ 0.6). **Furnishing does not change price.**

Q15. How does price per sq ft vary by property facing direction?

No meaningful difference (medians ≈ ₹9,220–9,310, p ≈ 0.35). Facing (Vastu) carries no price premium here.



Q11. Raw features are uncorrelated; the structure lives in the engineered features and targets

Investment, amenities and ownership (Q16–Q20)

Q16. How many properties belong to each owner type?

An almost exact three-way split: **Broker 83,479, Owner 83,268, Builder 83,253** (≈ 33.3% each).

Q17. How many properties are available under each availability status?

125,035 under construction and 124,965 ready to move (50/50). The raw file has no 'Sold' status.

Q18. Does parking space affect property price?

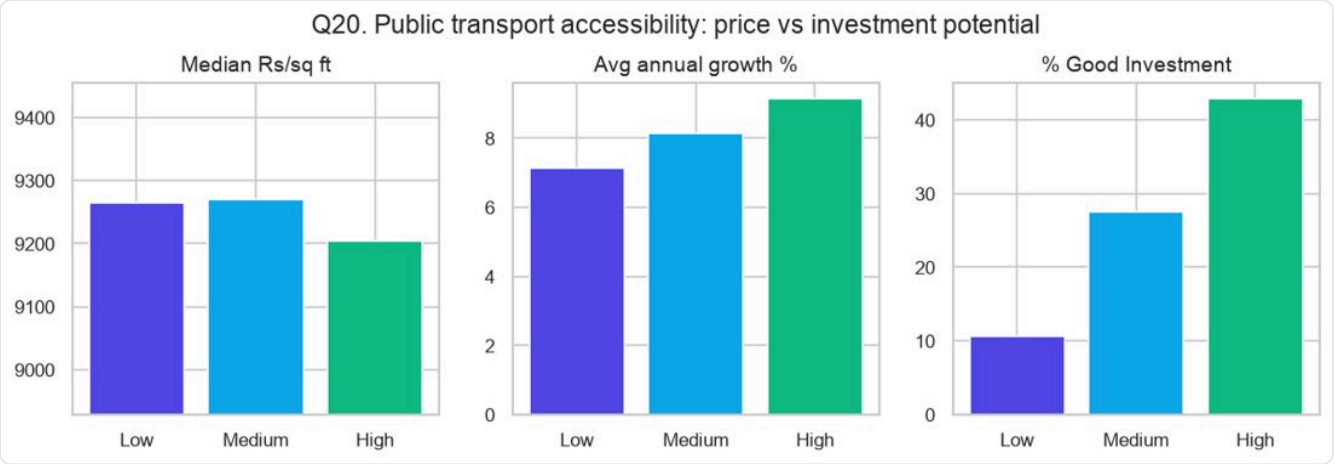
No. Average price is ₹254.4 L without parking and ₹254.8 L with it (Welch t-test p ≈ 0.58).

Q19. How do amenities affect price per sq ft?

Barely at all. Median price per sq ft is ≈ ₹9,200–9,300 for 1 to 5 amenities, and having any one amenity moves the median by less than ₹50 (< 0.5%). Like schools and hospitals, amenities matter for **appreciation**, not today's price.

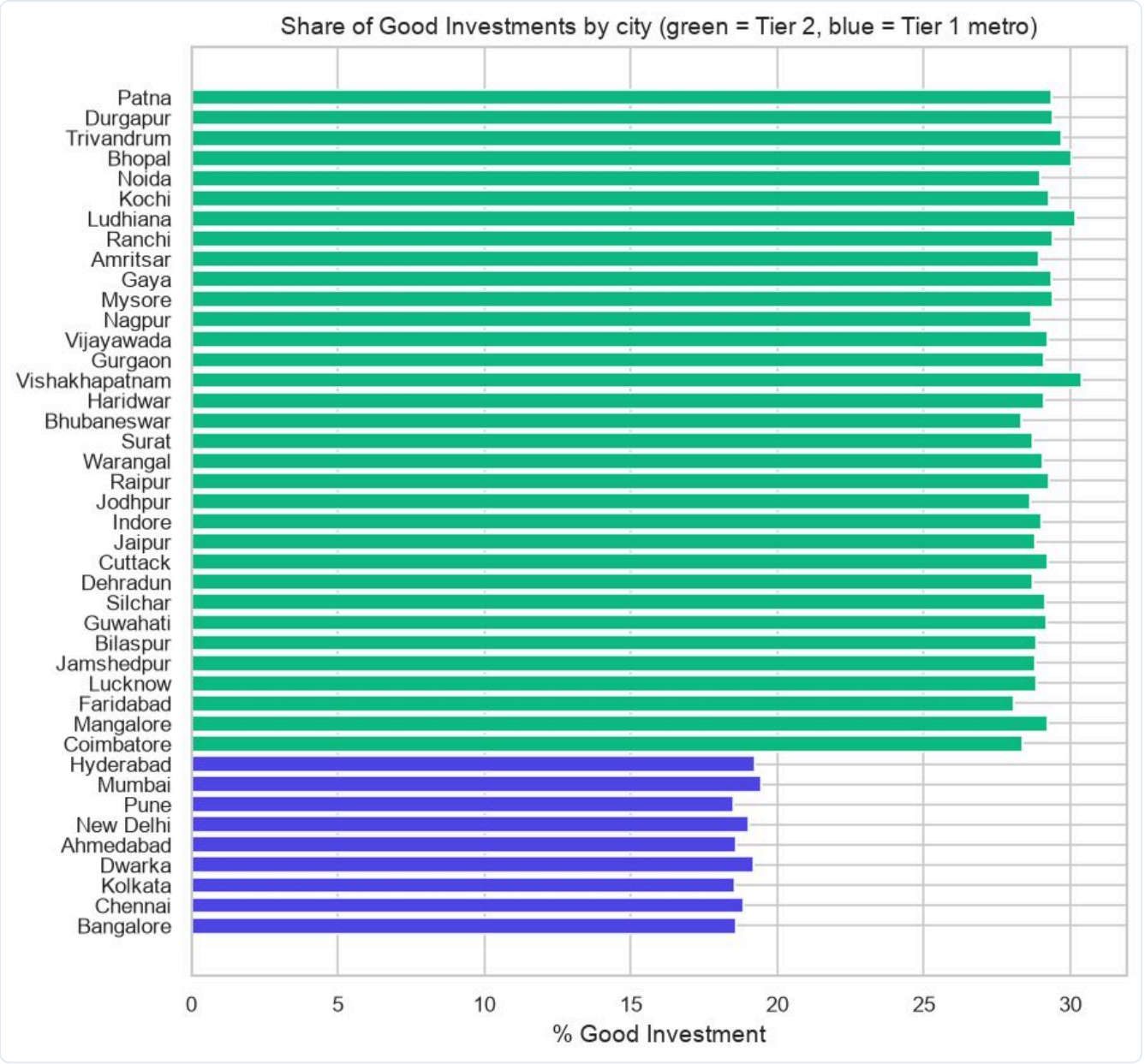
Q20. How does public transport accessibility relate to price per sq ft or investment potential?

Price per sq ft is the same at every level (≈ ₹9,200–9,270), but **investment potential is very different**: average growth climbs from 7.1% (low) to 8.1% (medium) to 9.1% (high), and the share of Good Investments from **10.7% to 27.5% to 42.9%**. Transport is the strongest single driver of the classification.

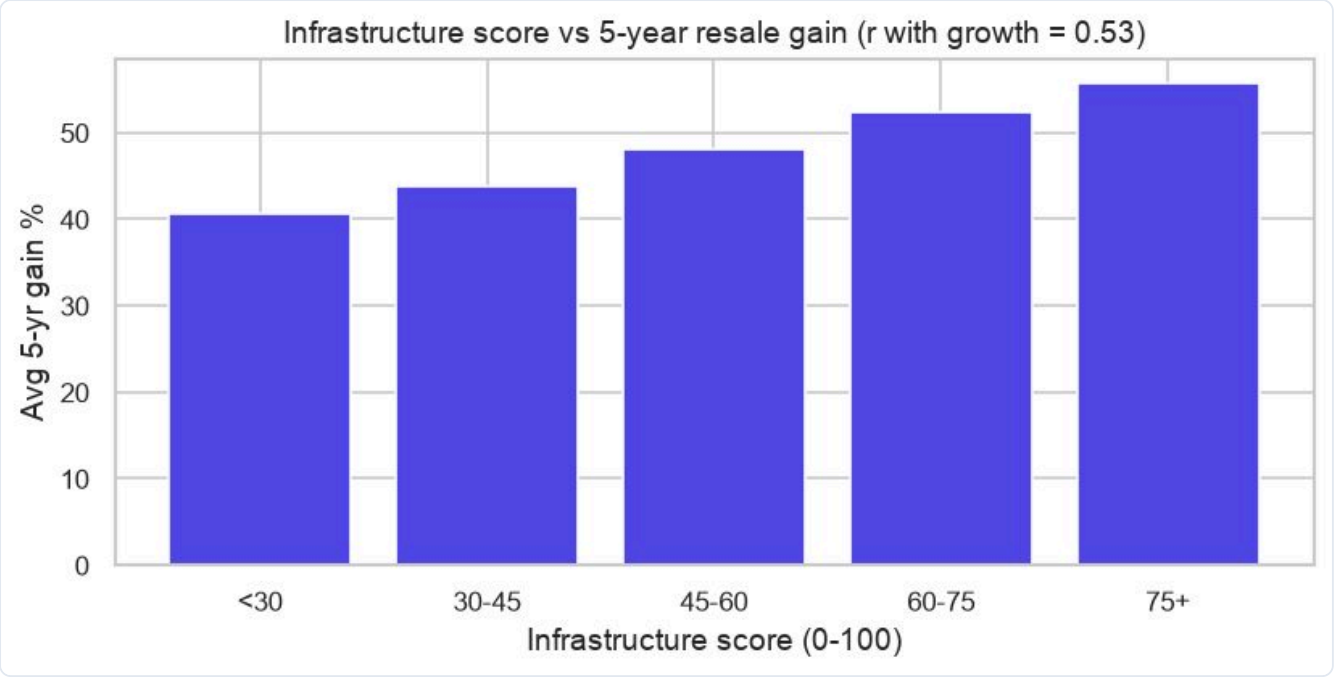


Q20. Transport does not change price per sq ft but strongly changes investment potential

Investment-focused analyses



Share of Good Investments by city: Tier 2 (green) ahead of Tier 1 metros (blue)



Infrastructure score vs expected 5-year resale gain

The dataset has **no crime-rate column**; the closest proxy, security, raises the Good Investment share from 25.3% to 28.8%.

Statistical tests

- **Chi-square test of independence** (categorical features vs Good_Investment) with Cramér's V for effect size: transport ($V \approx 0.30$), availability (0.12), city tier (0.09) and property type (0.07) are significant; owner type, furnishing and facing are not ($p > 0.7$).
- **Welch t-test**: under-construction properties grow significantly faster than ready-to-move ones.
- **One-way ANOVA**: growth differs significantly across transport levels.
- **Kruskal-Wallis** (non-parametric, skewed data): price per sq ft does **not** differ across the 42 cities ($p \approx 0.66$).
- **Pearson**: size vs price $r \approx 0.00$ ($p \approx 0.2$).

With 250,000 rows, tiny differences become "significant", so effect sizes were read alongside p-values (e.g. property type changes price per sq ft by ~1%: significant but not meaningful).

5. Model development

- One **stratified 70/15/15 split**, shared by both tasks: 175,000 train / 37,500 validation / 37,500 test. Models are fitted on train; **validation** picks the champions, each classifier's decision threshold and the forecast range; the **test** set is scored once and is the only set reported.
- A scikit-learn **Pipeline**: median/mode imputation → StandardScaler (numeric) + OneHotEncoder (categorical) → model, so the app applies exactly the same preprocessing.
- **Class imbalance** (73:27): stratified split, `class_weight="balanced"`, XGBoost `scale_pos_weight` = 2.70, and each classifier's probability cut-off tuned on validation to maximise F1. Models are compared on validation ROC AUC with F1 as tie-break (accuracy is misleading: the majority-class baseline already scores 73%).
- **Regression in log space**: price × $(1 + r)^5$ is multiplicative; log turns it additive, so every model fits it easily.
- **Hyperparameter tuning**: XGBoost, randomised search (12 candidates, 3-fold CV on a 40,000-row sample), logged as its own run.

Classification: Good Investment

Model	Val AUC	Test AUC	F1	Accuracy	Precision	Recall	Threshold
Baseline (majority class)	0.5000	0.5000	0.0000	0.7294	0.0000	0.0000	0.50
Logistic Regression	0.9462	0.9434	0.7739	0.8702	0.7319	0.8211	0.64
Decision Tree	0.9659	0.9640	0.8254	0.8982	0.7704	0.8889	0.62
Random Forest	0.9687	0.9668	0.8299	0.9017	0.7803	0.8863	0.65
Extra Trees	0.9575	0.9551	0.8115	0.8896	0.7544	0.8779	0.57
Hist Gradient Boosting	0.9717	0.9697	0.8363	0.9069	0.7972	0.8795	0.67
XGBoost	0.9719	0.9692	0.8339	0.9033	0.7793	0.8968	0.60
🏆 **XGBoost (tuned)**	**0.9722**	**0.9701**	**0.8381**	**0.9073**	**0.7950**	**0.8861**	**0.67**

Champion: XGBoost (tuned): best validation ROC AUC; on the test set ROC AUC 0.970, F1 0.838, accuracy 90.7%, precision 79.5%, recall 88.6% at a decision threshold of 0.67. Confusion matrix (test): 25,033 true negatives, 2,319 false positives, 1,156 false negatives, 8,992 true positives. Boosted trees lead because the label is an AND of two thresholds, which trees represent naturally; the remaining errors sit near the 8% growth line, where the random market shock decides the label.

Regression: price in 5 years

Model	Val RMSE	Test RMSE (L)	MAE (L)	MSE	R^2	MAPE %
Baseline (fixed 8% rate)	27.370	27.310	19.181	745.86	0.98328	5.09
🏆 **Linear Regression**	**14.955**	**14.904**	**10.388**	**222.14**	**0.99502**	**2.78**
Ridge Regression	14.955	14.904	10.388	222.13	0.99502	2.78
Decision Tree	18.221	18.122	12.742	328.41	0.99264	3.50
Random Forest	17.128	17.008	11.946	289.29	0.99352	3.29
Hist Gradient Boosting	15.155	15.078	10.531	227.35	0.99490	2.88
XGBoost	15.153	15.082	10.537	227.47	0.99490	2.88
XGBoost (tuned)	15.114	15.034	10.507	226.03	0.99493	2.87

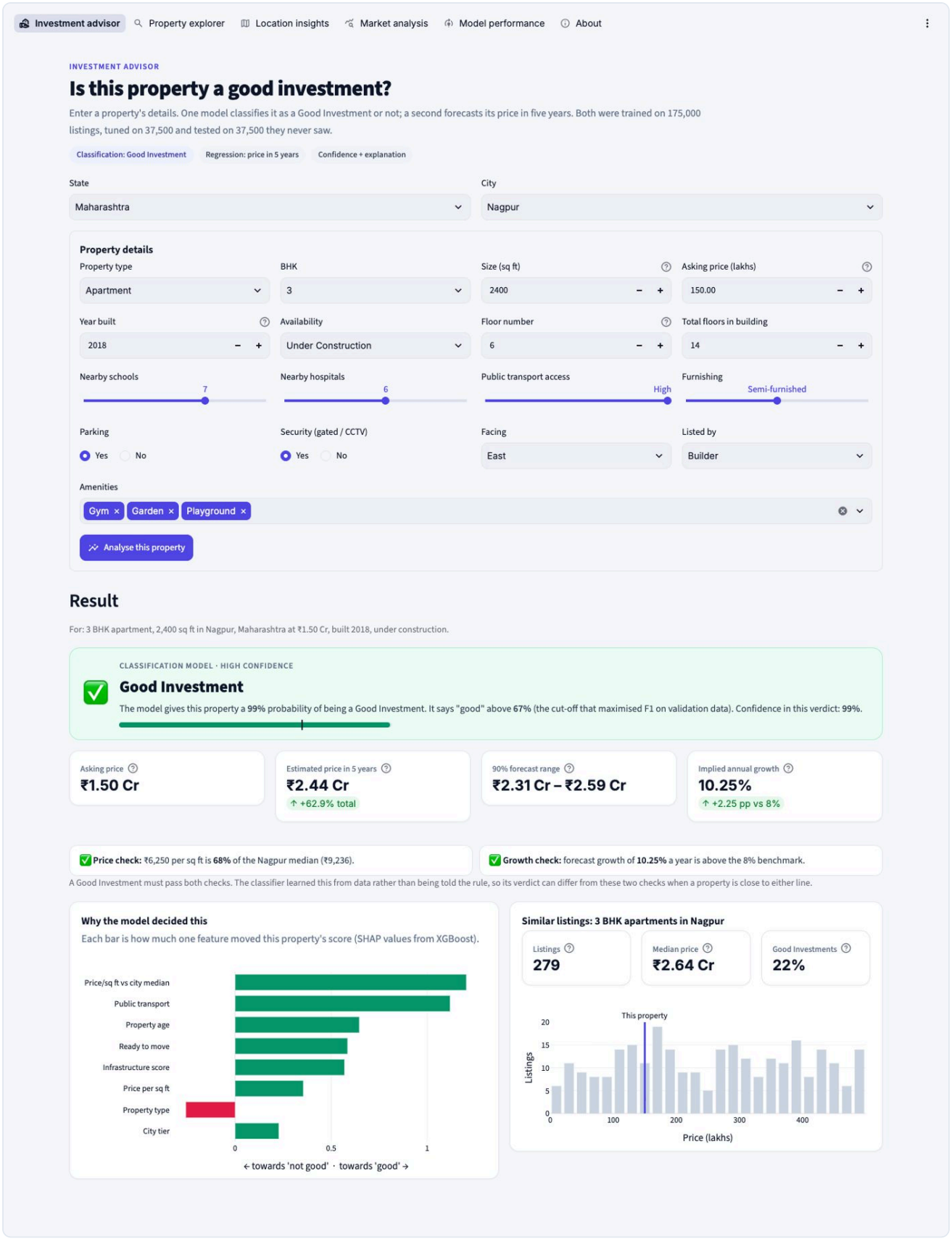
Champion: Linear Regression, RMSE 14.90 L (45% lower than the brief's fixed-8% formula) and MAPE 2.78%. In log space future price is a linear function of the features, so the simplest model wins and is also the fastest and most explainable. R^2 is high for every model because today's price explains most of the future price; RMSE against the baseline shows the real gain. The remaining error is the random market shock, which no model can predict. The app's 90% forecast range is calibrated on validation residuals and contains the true value for 89.8% of test listings.

6. MLflow tracking and registry

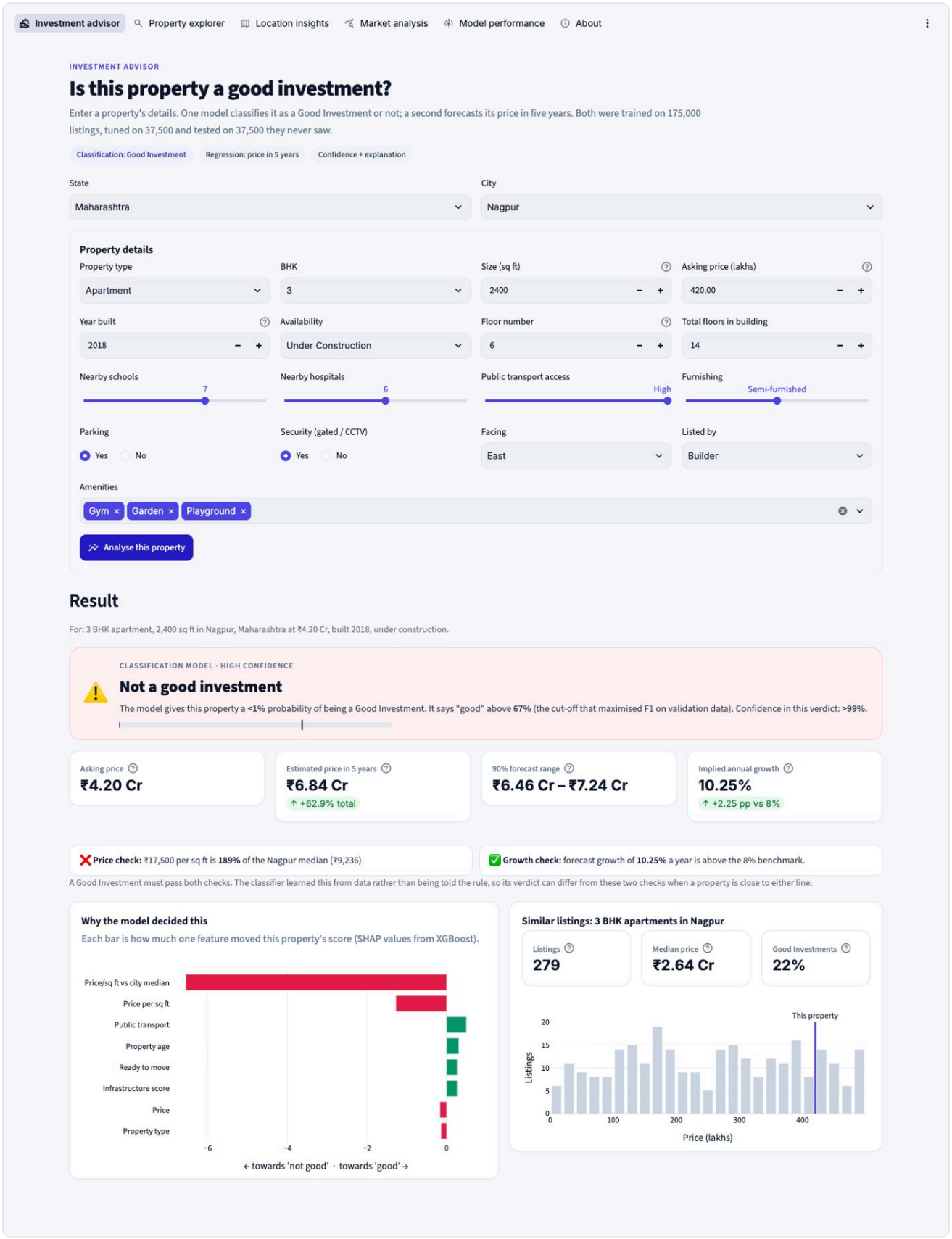
- Two experiments: `good_investment_classification` and `future_price_regression`.
- Every run logs hyperparameters, all metrics, a confusion matrix or actual-vs-predicted plot, a feature-importance chart and JSON, and the fitted pipeline with its input signature.
- Registry: `RealEstate_GoodInvestment_Classifier` v1 and `RealEstate_FuturePrice5Y_Regressor` v1, each with the alias `@champion`. The app loads these champions (exported to `models/`).
- `python evaluate.py` reloads both champions from the registry (`models:/…@champion`), re-scores the test set, prints the classification report and confusion matrix, and asserts that the numbers and the app's model files match exactly.
- `python mlflow_ui.py` opens the MLflow UI; it first re-points MLflow's stored artifact paths so a fresh clone works anywhere.

7. Streamlit application

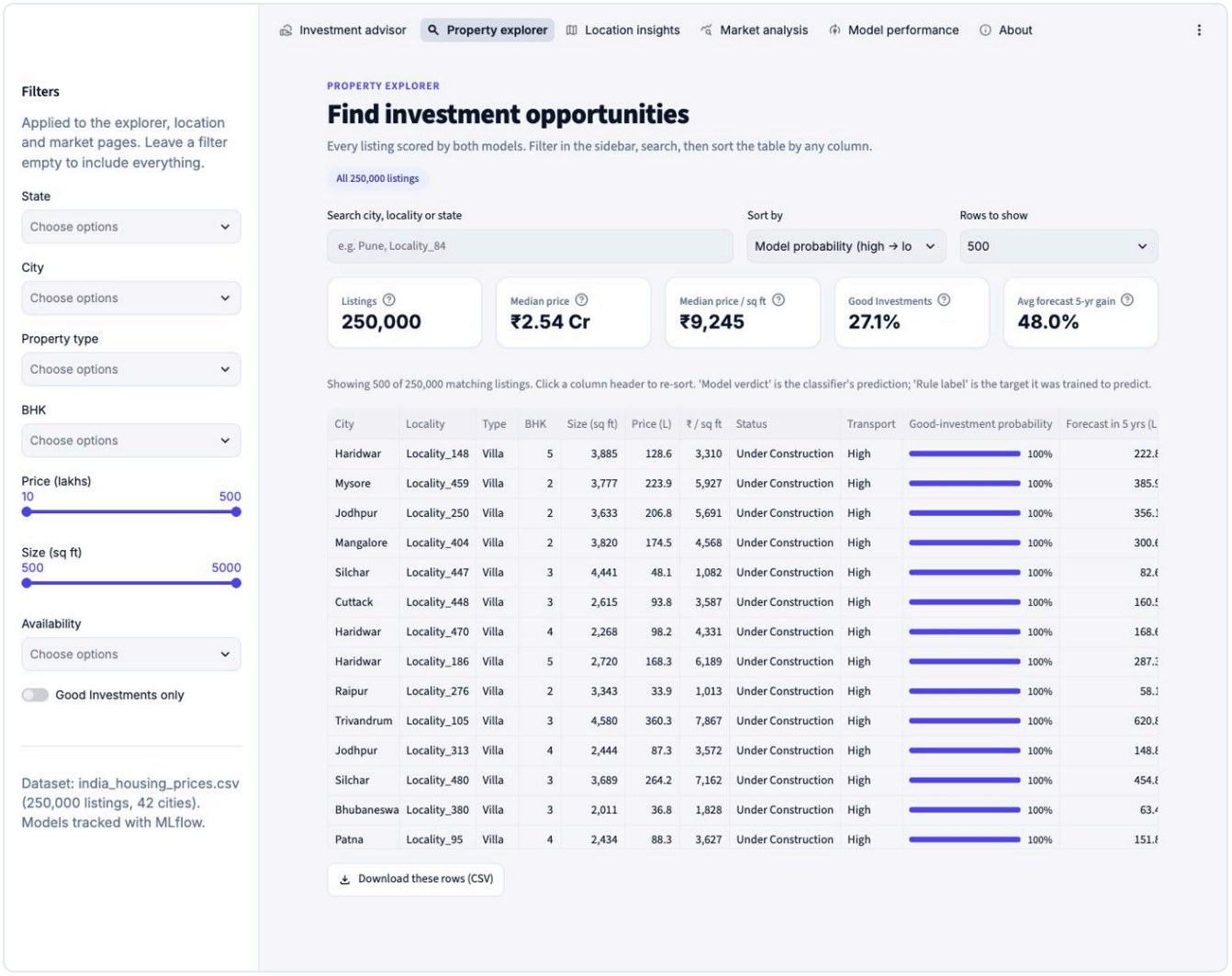
Six pages: **Investment advisor** (form → verdict with confidence, 5-year forecast with a 90% range, the two rule checks, SHAP explanation, similar listings), **Property explorer** (all 250,000 listings scored; filter by area, price, BHK, availability; search; sort; CSV download), **Location insights** (map heatmap, state × type heatmap, city ranking, price trend), **Market analysis**, **Model performance** and **About**.



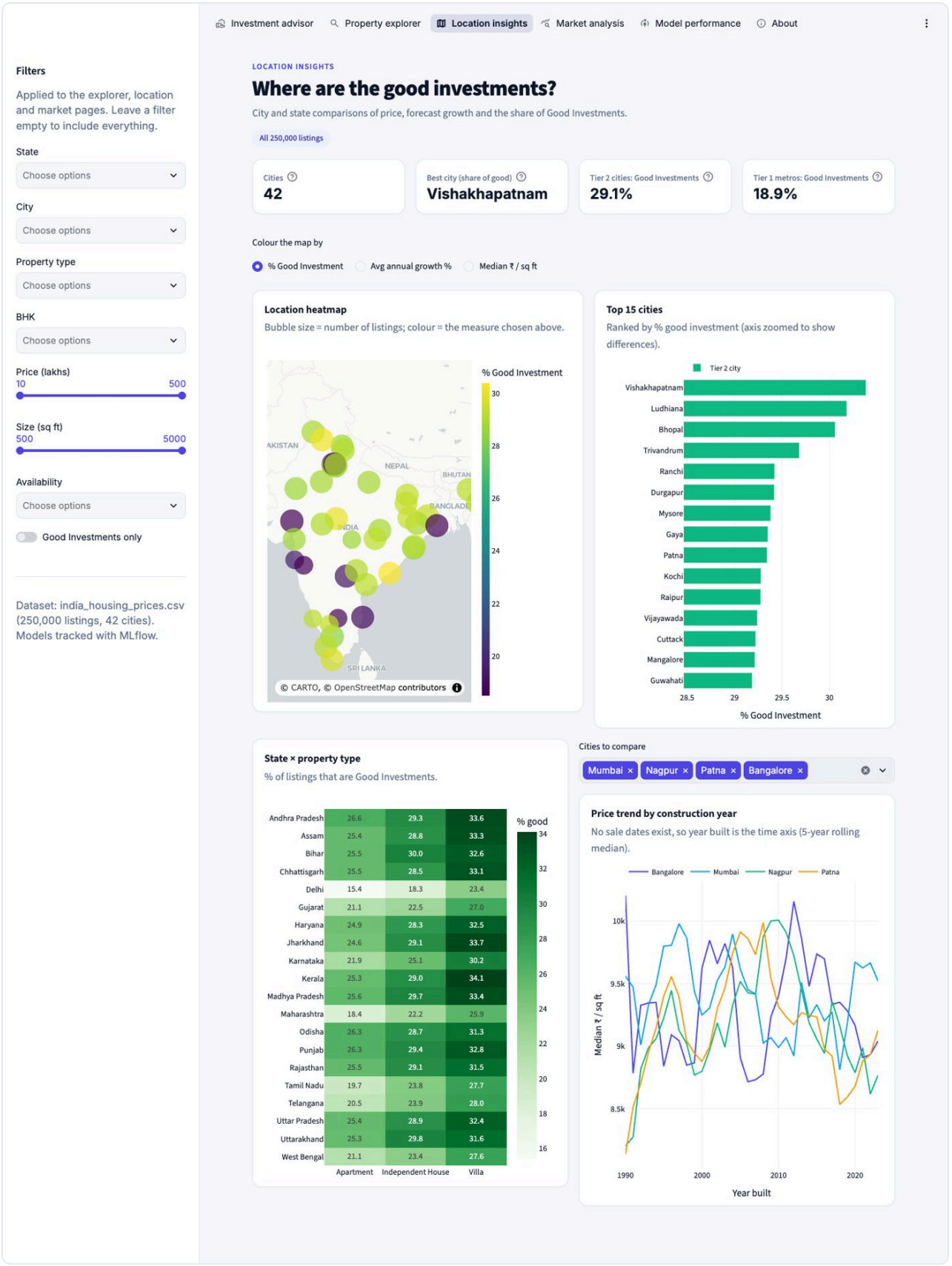
Investment advisor: a Good Investment with its forecast, checks and explanation



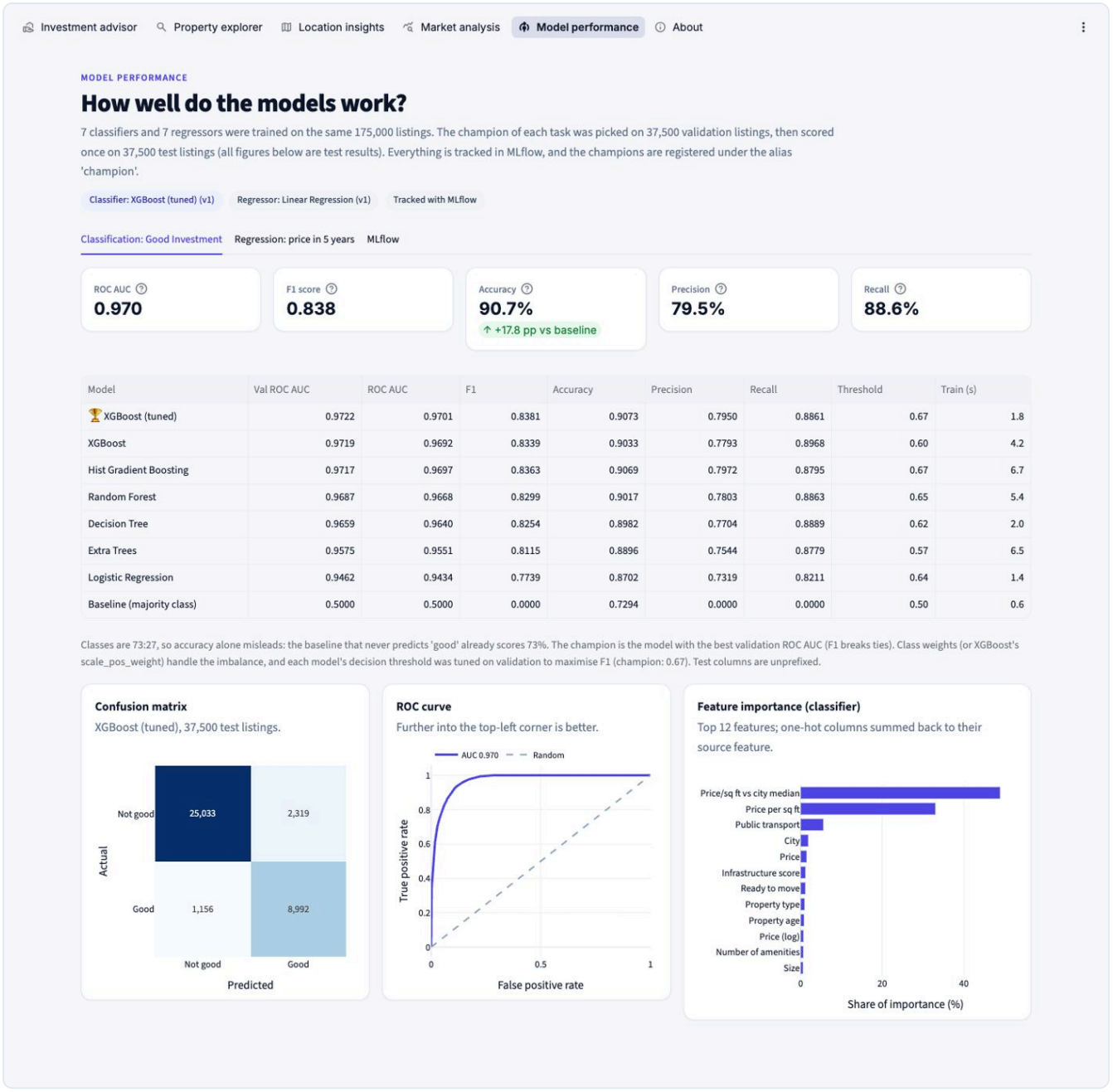
The same property at ₹4.2 Cr fails the price check, and the verdict flips



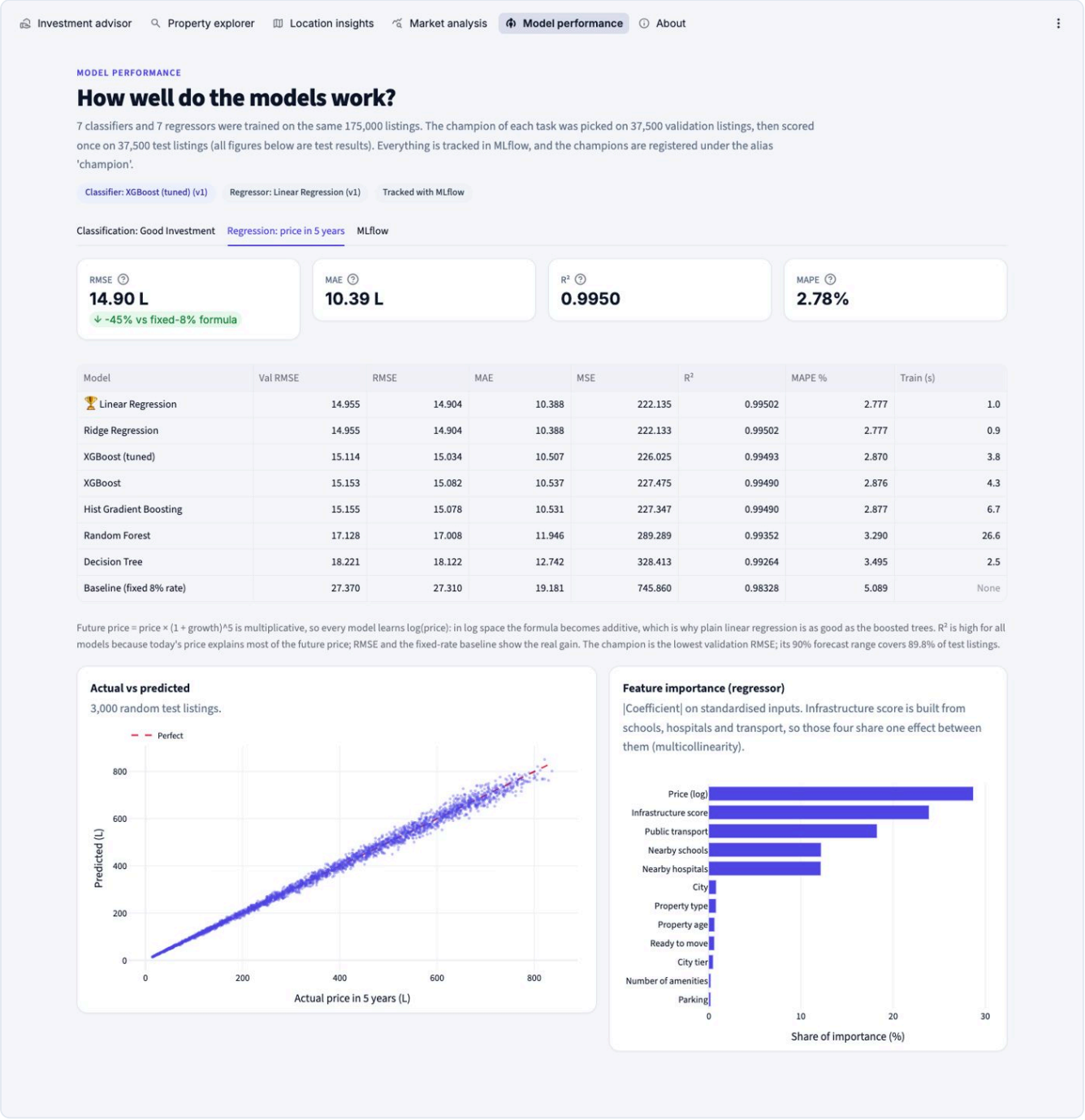
Property explorer: every listing scored by both models



Location insights: map heatmap, city ranking, state × property type, trends



Model performance: classification comparison, confusion matrix, ROC, importance



Model performance: regression comparison and actual vs predicted

8. Insights and business recommendations

1. **Connectivity first.** High-transport properties are Good Investments 43% of the time vs 11% for low transport. Platforms should surface transport access prominently, and investors should weight it above furnishing or facing.
2. **Buy the asset, not the finish.** Furnishing, facing and owner type show no relationship with price or returns.
3. **Look beyond the metros.** Tier 2 cities offer 29% Good Investments vs 19% in the eight metros, matching the "developing areas" use case.
4. **Price per sq ft matters more than price.** Larger homes are cheaper per sq ft and pass the value test far more often (45% Good Investments above 4,000 sq ft vs 7% below 1,000).

5. **Under-construction purchases** carry higher expected appreciation; pair them with developer due diligence (RERA).
6. **Use the model as a screen, not a verdict.** It ranks 250,000 listings in seconds; humans review the shortlist.

Limitations and next steps

- The listings are synthetic and the targets are rule-based, so the models show how well each algorithm recovers a documented investment rule, not real market behaviour.
- With real transactions (registration data, historical prices), the same pipeline would learn the growth drivers directly. Useful additions: crime index, RERA status, distance to metro and business districts.
- Monitoring: retrain on new data and compare against the registered champion in MLflow before promoting.

9. Deliverables

Cleaned dataset	`data/processed/housing_clean.csv`
Scripts	`pipeline.py` (data), `train.py` (training + MLflow), `evaluate.py` (evaluation), `src/` (cleaning, features, targets, EDA, modelling, tracking)
EDA notebook	`notebooks/eda.ipynb` (all 20 questions + statistical tests)
MLflow logs + registry	`mlflow.db` , `mlruns/` , viewed with `python mlflow_ui.py`
Streamlit app	`app/streamlit_app.py`
Documentation	`README.md` , this report, `reports/screenshots/`